

**Multiple Choice:**

1. What common connector usually joins RG-213 coaxial cable to an HF transceiver?
 - A. PL-259 connector
 - B. An F-type cable connector
 - C. A banana plug connector
 - D. A binding post connector
2. The characteristic impedance of a transmission line is determined by the:
 - A. length of the line
 - B. physical dimensions and relative positions of the conductors
 - C. frequency at which the line is operated
 - D. load placed on the line
3. A flexible coaxial line contains:
 - A. four or more conductors running parallel
 - B. only one conductor
 - C. braid and insulation around a central conductor
 - D. two parallel conductors separated by spacers
4. Where would you install a balun to feed a dipole antenna with 50-ohm coaxial cable?
 - A. Between the coaxial cable and the antenna
 - B. Between the transmitter and the coaxial cable
 - C. Between the antenna and the ground
 - D. Between the coaxial cable and the ground
5. The characteristic impedance of a parallel wire transmission line does not depend on the:
 - A. velocity of energy on the line
 - B. radius of the conductors
 - C. center to center distance between conductors
 - D. dielectric
6. What are some reasons not to use parallel-conductor feed line?
 - A. You must use an impedance-matching device with your transceiver, and it does not work very well with a high SWR
 - B. It does not work well when tied down to metal objects, and it cannot operate under high power
 - C. It does not work well when tied down to metal objects, and you must use an impedance-matching device with your transceiver
 - D. It is difficult to make at home, and it does not work very well with a high SWR
7. What kind of antenna feed line is made of two conductors held apart by insulated rods?
 - A. Open-conductor ladder line
 - B. Coaxial cable
 - C. Twin lead in a plastic ribbon
 - D. Twisted pair
8. What kind of antenna feed line can be constructed using two conductors which are maintained a uniform distance apart using insulated spreaders?
 - A. Coaxial cable
 - B. 75 ohm twin-lead
 - C. 600 ohm open-wire
 - D. 300 ohm twin-lead
9. What does the term "balun" mean?
 - A. Balanced unloader
 - B. Balanced to unbalanced
 - C. Balanced unmodulator
 - D. Balanced antenna network
10. The characteristic impedance of a transmission line is:
 - A. the impedance of a section of the line one wavelength long
 - B. the dynamic impedance of the line at the operating frequency
 - C. the ratio of the power supplied to the line to the power delivered to the termination
 - D. equal to the pure resistance which, if connected to the end of the line, will absorb all the power arriving along it
11. What factors determine the characteristic impedance of a coaxial antenna feed line?
 - A. The ratio of the diameter of the inner conductor to the diameter of the braid
 - B. The diameter of the braid and the length of the line
 - C. The diameter of the braid and the frequency of the signal
 - D. The frequency of the signal and the length of the line
12. What is a coaxial cable?
 - A. Two wires side-by-side in a plastic ribbon
 - B. Two wires side-by-side held apart by insulating rods
 - C. Two wires twisted around each other in a spiral
 - D. A center wire inside an insulating material which is covered by a metal sleeve or shield
13. A 75 ohm transmission line could be matched to the 300 ohm feedpoint of an antenna:
 - A. with an extra 250 ohm resistor
 - B. by using a 4 to 1 balun
 - C. by using a 4 to 1 trigatron
 - D. by inserting a diode in one leg of the antenna
14. What connects your transceiver to your antenna?
 - A. A ground wire
 - B. A dummy load
 - C. A transmission line
 - D. The power cord
15. What is parallel-conductor feed line?
 - A. Two wires twisted around each other in a spiral
 - B. A center wire inside an insulating material which is covered by a metal sleeve or shield
 - C. A metal pipe which is as wide or slightly wider than a wavelength of the signal it carries



- D. Two wires side-by-side held apart by insulating rods
16. What common connector usually joins a hand-held transceiver to its antenna?
- A. A BNC connector
 - B. A PL-259 connector
 - C. An F-type cable connector
 - D. A binding post connector
17. What is an unbalanced line?
- A. Feed line with neither conductor connected to ground
 - B. Feed line with both conductors connected to ground
 - C. Feed line with both conductors connected to each other
 - D. Feed line with one conductor connected to ground
18. Which of these common connectors has the lowest loss at UHF?
- A. An F-type cable connector
 - B. A BNC connector
 - C. A PL-259 connector
 - D. A type-N connector
19. What factors determine the characteristic impedance of a parallel-conductor antenna feed line?
- A. The distance between the centres of the conductors and the radius of the conductors
 - B. The distance between the centres of the conductors and the length of the line
 - C. The radius of the conductors and the frequency of the signal
 - D. The frequency of the signal and the length of the line
20. What commonly available antenna feed line can be buried directly in the ground for some distance without adverse effects?
- A. 300 ohm twin-lead
 - B. 600 ohm open-wire
 - C. 75 ohm twin-lead
 - D. coaxial cable
21. Why does coaxial cable make a good antenna feed line?
- A. It is weatherproof, and its impedance is higher than that of most amateur antennas
 - B. It is weatherproof, and its impedance matches most amateur antennas
 - C. It can be used near metal objects, and its impedance is higher than that of most amateur antennas
 - D. You can make it at home, and its impedance matches most amateur antennas
22. What device can be installed to feed a balanced antenna with an unbalanced feed line?
- A. A triaxial transformer
 - B. A balun
 - C. A wavetrap
 - D. A loading coil
23. What is the best antenna feed line to use, if it must be put near grounded metal objects?
- A. Ladder-line
 - B. Twisted pair
 - C. Coaxial cable
 - D. Twin lead
24. A transmission line differs from an ordinary circuit or network in communications or signaling devices in one very important way. That important aspect is:
- A. capacitive reactance
 - B. inductive reactance
 - C. propagation delay
 - D. resistance
25. The impedance of a coaxial line:
- A. can be the same for different diameter line
 - B. changes with the frequency of the energy it carries
 - C. is correct for only one size of line
 - D. is greater for larger diameter line

END